

UNIT – IV: TESTING OF HYPOTHESIS

High Priority Question Set (Final Preparation List)

SECTION A: THEORY

1. Define the following:
 - i) Null Hypothesis and Alternative Hypothesis
 - ii) Type I and Type II errors
 - iii) Level of Significance
 2. Explain the procedure generally followed in testing of hypothesis.
 3. Explain one-tailed and two-tailed tests with suitable examples.
 4. Write the test statistics for:
 - i) Single mean
 - ii) Single proportion
 - iii) Difference of means
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SECTION B: LARGE SAMPLE TESTS (Z-TEST)

5. A sample of 100 students has a mean weight of 50 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 16 kgs? Test at 1% level of significance.
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6. A sample of 400 items is taken from a population whose standard deviation is 10. The mean of the sample is 40. Test whether the sample has come from a population with mean 38.
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7. In a random sample of 60 workers, the average time taken is 33.8 minutes with a standard deviation of 6.1 minutes. Test the hypothesis $\mu = 32.6$ against $\mu > 32.6$ at 1% level of significance.
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8. The means of two large samples of sizes 1000 and 2000 are 67.5 inches and 68.0 inches respectively. Test whether the samples are drawn from the same population with standard deviation 2.5 inches.
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SECTION C: PROPORTION TESTS

9. In a big city, 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?
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10. In a study, 63 out of 100 urban residents and 59 out of 125 suburban residents support the construction of a nuclear power plant. Test whether there is a significant difference between the two proportions at 5% level of significance.

11. A machine A produced 20 defective articles in a batch of 400 and machine B produced 10 defectives in a batch of 300. Test whether the two machines have the same efficiency at 5% level of significance.

12. Experience shows that 20% of manufactured products are of top quality. In a sample of 400 articles, only 50 are of top quality. Test the hypothesis at 5% level of significance.

13. In a sample of 120 cool drinkers, 68 prefer Thums Up. Test the null hypothesis $p = 0.5$.

SECTION D: SMALL SAMPLE TESTS (t-TEST)

14. A sample of 10 bolts has mean length 0.53 mm and standard deviation 0.03 mm. Test whether the machine is in proper working condition if the standard length is 0.50 mm at 5% level of significance.

15. The following observations represent changes in blood pressure after administering a medicine:
7, 3, -1, 4, -3, 5, 6, -4, -1
Test whether the medicine increases blood pressure at 5% level of significance.

16. To examine the hypothesis that husbands are more intelligent than wives, the following IQ scores were recorded:

Husbands: 117, 105, 97, 105, 123, 109, 86, 78, 103, 107

Wives: 106, 98, 87, 104, 116, 95, 90, 69, 108, 85

Test the hypothesis at 0.05 level of significance.

17. The following data represent the gain in weights (in kgs) of pigs fed on two diets A and B:

Diet A: 25, 32, 30, 34, 24, 14, 32, 24, 30, 31, 35, 25

Diet B: 44, 34, 22, 10, 47, 31, 40, 30, 32, 35, 18, 21, 35, 29, 22

Test whether the two diets differ significantly in their effect on weight gain.

SECTION E: CHI-SQUARE TESTS

18. A die is thrown 100 times with the following results. Test whether the die is unbiased at 5% level of significance.

No. on die	1	2	3	4	5	6
Frequency	10	12	24	16	20	18

19. Four coins were tossed 160 times and the number of heads obtained is given below. Test the goodness of fit at 5% level of significance.

No. of Heads	0	1	2	3	4
Frequency	19	54	58	23	6

SECTION F: F-TEST (VARIANCE TEST)

20. The time taken by workers using two different methods is given below:

Method-1	20	16	26	27	23	22
Method-2	27	33	42	35	32	34

Test whether the variances of the two methods differ significantly. (Tabulated value = 4.39)